Water quality monitoring system

"we are building water quality checker with camera using ph strip and then using app rating water tankers coming area by area" *Generated by LeapSpace*

Yes — this is a **feasible water-quality monitoring system**, and the strongest design pattern in the evidence is: **camera + pH strip/image analysis + mobile app + location tagging + area-level aggregation/rating** for water tanker monitoring and mapping. Smartphone- and camera-based colorimetric pH systems have repeatedly shown that image processing can replace subjective visual reading and produce quantitative results, while mobile apps can support on-device analysis and field use [1] [2] [3] [4] [5].

What your system should look like

Module	Recommended function	Why it fits your idea
Sampling + pH strip	User dips strip into tanker water	Colorimetric strips are widely used for low-cost field measurements, but visual reading is subjective [1] [6].
Camera capture	Phone camera captures strip image	Smartphone camera capture is a proven input for pH and other water-quality assays [3] [5] [7].
Image correction	White balance, crop, ROI detection, lighting normalization	Lighting and camera variation are known problems; several systems reduce this with calibration, reference charts, or controlled capture conditions [2] [4] [5] [8].
pH estimation model	Convert color features to pH using ML or calibrated color-distance rules	Both machine-learning and color-distance approaches have been used successfully for pH strip quantification [2] [3] [9] [10] [11].
App scoring layer	Convert pH and other checks into a tanker rating	Water-quality apps can already transform field readings into user-facing outputs, maps, and management decisions [12] [13] [14] [15].
Area mapping	Store GPS and aggregate by neighborhood/zone	GPS-tagging and mapping are established in water-quality citizen-science systems [12] [13].

Best implementation choices

1) Use controlled image capture

A mini light box, fixed-distance capture, or reference card will improve consistency because ambient light and camera auto-correction can distort strip colors [4] [5] [8] [12]. If you want a simpler low-cost approach, a web/app pipeline that extracts RGB from the strip and compares it to reference values can work without heavy ML [11].

2) Start with pH only, then expand

The literature shows pH is one of the most mature use cases for strip imaging, with systems ranging from direct RGB analysis to ML-based Android apps [1] [2] [7] [9]. Once the pH pipeline is stable, you can add other strip-based water indicators later, as demonstrated in multi-parameter paper-based platforms [16] [17].

3) For tanker rating, combine measurement + geography + time

The evidence supports mapping water-quality results and using them for area-level monitoring rather than only single measurements [12] [13] [14]. That means your app can rank tankers by:

- **current pH result**
- **repeatability over time**
- **location / area**
- **trend compared with prior samples**
  This is an inference from the cited monitoring systems' mapping and repeated-sampling logic [12] [13] [15].

Practical recommendation

The most supportable first version is:

1. **User scans pH strip with phone camera** [1] [5].
2. **App standardizes image** using crop + white balance + reference calibration [4] [10] [11].
3. **Model estimates pH** using either calibrated color matching or ML [2] [3] [9].
4. **App stores GPS + timestamp** and uploads to a backend map [12] [13].
5. **Backend computes tanker/area rating** from pH and repeated observations, then shows hotspot maps and low-quality alerts [14] [15].

Bottom line

Your idea is technically aligned with existing water-quality research, but the key to making it reliable is **standardized photo capture and a calibrated scoring model**, not just raw camera color reading [4] [5] [8]. If you design it as a **field monitoring + mapping platform** rather than only a one-off pH reader, it matches the strongest evidence base for scalable, low-cost water-quality assessment [12] [13] [15].

Confidence level: **High** — multiple independent studies support smartphone/camera-based colorimetric water testing, image normalization, and location-based water-quality monitoring, though tanker-specific rating logic is an inference built from these established patterns [1] [4] [5] [11] [12] [13].

References

1. Ding, Y., Wang, Q. (2011) A rapid quantitative detection method of colorimetric test strip based on machine vision. *Journal of Food, Agriculture and Environment*. https://www.scopus.com/pages/publications/80655133808
2. Yıldız, E., Şen, M., Ozdemir, M.A. (2026) Smartphone-based colorimetric analysis of pH strips using machine learning. *Analytical Methods*. https://doi.org/10.1039/d6ay00780e
3. Mutlu, A.Y., Kılıç, V., Özdemir, G.K., (...), Solmaz, M.E. (2017) Smartphone-based colorimetric detection: Via machine learning. *Analyst*. https://doi.org/10.1039/c7an00741h
4. Kim, S.D., Koo, Y., Yun, Y. (2017) A smartphone-based automatic measurement method for colorimetric pH detection using a color adaptation algorithm. *Sensors (Switzerland)*. https://doi.org/10.3390/s17071604
5. Kılıç, V., Alankus, G., Horzum, N., (...), Solmaz, M.E. (2018) Single-Image-Referenced Colorimetric Water Quality Detection Using a Smartphone. *ACS Omega*. https://doi.org/10.1021/acsomega.8b00625
6. Vizárová, K., Vajová, I., Krivoňáková, N., (...), Katuščák, S. (2021) Regression Analysis of Orthogonal, Cylindrical and Multivariable Color Parameters for Colorimetric Surface pH Measurement of Materials. *Molecules (Basel, Switzerland)*. https://doi.org/10.3390/molecules26123682
7. Heidari, S., Mirzaee – Ghaleh, E., Rabbani, H., Vesali, F. (2021) Development of an android app for estimating the water quality parameters in fish pond. *Environmental Science and Pollution Research*. https://doi.org/10.1007/s11356-021-12974-y

8. Jantarakasem, C., Sioné, L., Templeton, M.R. (2026) A critical review of the use of smartphone cameras in water quality analysis. *Environmental Technology Reviews*. https://doi.org/10.1080/21622515.2025.2593442
9. Bursalı, E.M., Ozdemir, M.A., Şen, M. (2025) High-accuracy machine learning–based colorimetric pH quantification using a custom-built portable strip-imaging device and smartphone application. *Microchimica Acta*. https://doi.org/10.1007/s00604-025-07682-0
10. Argade, M., Bharat, S., Chopra, P.K.P.G., (...), Jichkar, A.A. (2025) Colorimetric pH Detector Using Python. *2025 3rd DMIHER International Conference on Artificial Intelligence in Healthcare, Education and Industry, IDICAIHEI 2025*. https://doi.org/10.1109/IDICAIHEI65991.2025.11378805
11. McCarty, D.A., Kim, M.A., Jo, H., (...), Kwon, S. (2025) Accessible water quality monitoring through hybrid human–machine colorimetric methods. *Environmental Monitoring and Assessment*. https://doi.org/10.1007/s10661-025-13983-x
12. Sicard, C., Glen, C., Aubie, B., (...), Filipe, C.D.M. (2015) Tools for water quality monitoring and mapping using paper-based sensors and cell phones. *Water Research*. https://doi.org/10.1016/j.watres.2014.12.005
13. Li, J., Gao, M., Zhang, B., (...), Xie, Y. (2022) Smartphone photo based water quality monitoring algorithm and application. *Cehui Xuebao/Acta Geodaetica et Cartographica Sinica*. https://doi.org/10.11947/j.AGCS.2022.20220049
14. Santosa, A.F., Huang, Y., Ashlah, M.B., (...), Sean, W.-Y. (2025) Analysis of Reservoir Water Quality by Smartphone Color Image Analysis: A Case Study of Three Reservoirs in Taiwan. *Applied Sciences (Switzerland)*. https://doi.org/10.3390/app152312370
15. Jakositz, S., Pillsbury, L., Greenwood, S., (...), Mo, W. (2020) Protection through participation: Crowdsourced tap water quality monitoring for enhanced public health. *Water Research*. https://doi.org/10.1016/j.watres.2019.115209
16. Dawan, P., Chaiyasith, W.C., Sawangmek, S. (2026) Development of a paper-based analytical device integrated with a custom-built smartphone application for on-site colorimetric water quality testing. *Talanta*. https://doi.org/10.1016/j.talanta.2025.128957
17. Xiong, X., Guo, C., Yan, G., (...), Han, J. (2022) Simultaneous Cross-type Detection of Water Quality Indexes via a Smartphone-App Integrated Microfluidic Paper-Based Platform. *ACS Omega*. https://doi.org/10.1021/acsomega.2c05938